

IRNSS: India's Attempt At Gaining INDEPENDENCE FROM GPS



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Indian Regional Navigation Spacecraft System (IRNSS) is the Indian navigation system that envisages the establishment of the regional navigation system using a combination of geostationary orbit (GEO) and geosynchronous orbit (GSO) spacecraft,” reads Indian Space Research Organisation (ISRO) website on self-reliance with an indigenous navigation system. What does this mean for the common man?

We open Google Maps application on our phones and search for a place. The signal travels all around the world, to the GPS control station in the USA. Processing takes place at their end and, after some deliberation on their part, we receive certain co-ordinates. This takes some time.

Also, according to conspiracy theorists, security agencies get to know if you access anything disconcerting. With IRNSS, the time to access is expected to be much lesser and the said theorists can feel a bit safer, as this information will now be processed by Indian agencies.

Portable GNSS simulator



Facts and figures

Budget: ₹ 14.4 billion

Coverage: 1500km from Indian boundary

Control centre: ISRO Navigation Centre at Bhyaluru near Bengaluru

Offered services: Standard Positioning Service (public users) and Restricted Service (authorised users)

Project timeline

“It was back in 2003 that the project was beginning to take form,” recalls Dr Vyasaraj G., senior manager - GNSS and aerospace electronics, Accord Software. “The main focus behind developing an indigenous system was to remove dependence on the American GPS system. After announcement of the project we were to design simulators that could simulate the perfect environments for testing the satellites, and this began the race for development of IRNSS receiver modules.

“After finalisation of the systems, we had receiver modules being developed to test signal characteristics before the first satellite was in place. By the end of 2012, we had developed receiver modules, even before the satellites were launched,” adds Dr Vyasaraj.

By 2008, project details were finalised and the expected date for the system to go live was set to end of 2015, with the first satellite being launched in 2011, and the next every six months. After a delay in the original timeline, the first satellite, IRNSS-1A, was launched on July 1, 2013, and the others followed suit. The last satellite, IRNSS-1G, was successfully launched on April 28, 2016, from Satish Dhawan Space Centre (SDSC), SHAR, Sriharikota launch site.

Development and deployment

The proposed system consists of a constellation of seven satellites and a support